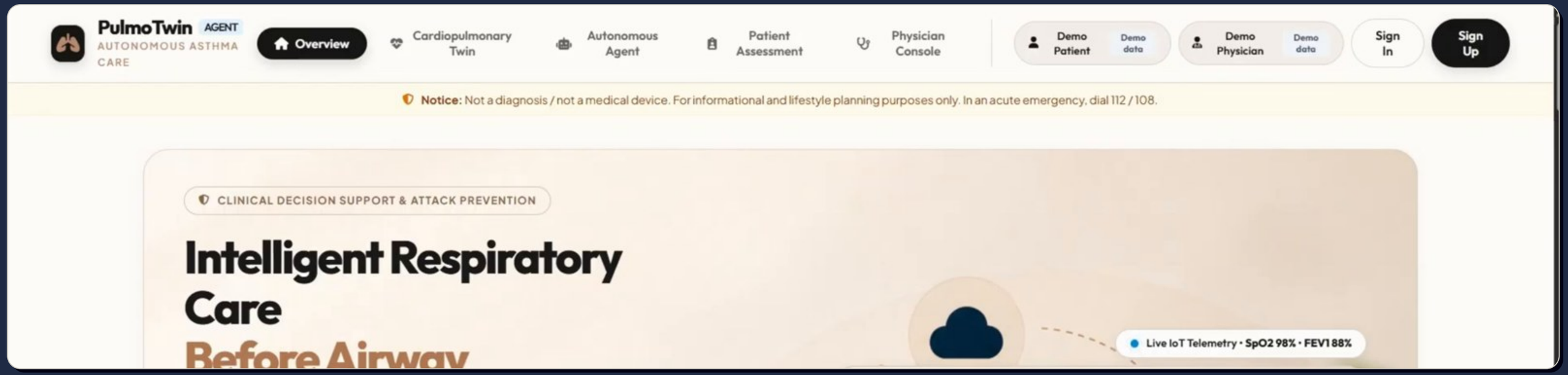
# PulmoTwin

## Multimodal Cardiopulmonary Digital Twin

for Proactive Adverse Event Forecasting and Personalised Clinical Intervention

*Pairing continuous wearables, ABDM electronic health records and atmospheric telemetry*

**Digital Twin Challenge 2026** | **Happiest Health | Reimagining & Reforming Healthcare in India Summit 2026** | **Bengaluru, 20 Nov 2026**




**Team PulmoTwin**

Team lead: Kabir Roy

Live cockpit: kabirroy12345.github.io/Digital_Twin

Code (MIT): github.com/Kabirroy12345/Digital_Twin

Demo video: youtube.com/watch?v=i7NJstq8_ZE

**RESEARCH PROTOTYPE, NOT A MEDICAL DEVICE**

Investigational decision-support tool. Not cleared by CDSCO, FDA or CE. It does not diagnose or change prescriptions. In an emergency dial 112 / 108.

THE PROBLEM

# India's asthma care is reactive, late and invisible

**37.9M**
people in India live with bronchial asthma

**42%**
of global asthma deaths, with 13% of asthmatic patients

**300-500%**
surge in airway resistance before chest tightness is felt

**>80%**
of patients cannot recall 30 days of symptoms and puffs

### The reactive care trap

- Patients check air quality only after wheezing starts. By then airway resistance has already surged, and the next step is emergency nebulisation or the ICU.

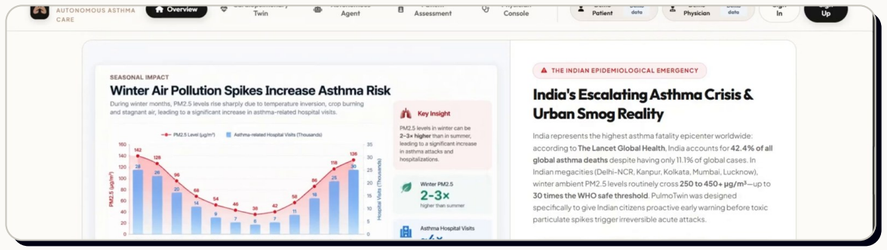
### Deceptive winter inversions

- In Delhi, Mumbai and Bengaluru a clear morning sky hides trapped PM2.5. Patients skip maintenance inhalers or jog outdoors right before the spike.

### The clinical memory void

- GINA 2023 flags reliever use above 2 times a week as poor control, yet a 5-minute consultation cannot reconstruct a 30-day symptom and puff history.

### What the product shows about the burden




*Landing page: India's escalating asthma crisis (demo video)*

Source figures: Lancet Global Health, GINA 2023 guidance, as cited in the project README.

### The gap PulmoTwin closes

- Moves care from react-after-symptoms to act-hours-before
- Turns invisible risk (PM2.5, HRV, airway load) into a forecast the patient can act on
- Gives the physician an objective 30-day record instead of memory

THE SOLUTION

# A living virtual replica of the patient's lungs and autonomic state

PulmoTwin does not compute a static risk score. It continuously ingests three streams, simulates the patient's physiology forward 24 hours, and tests interventions in silico before anything is done in real life.

### Longitudinal EHR

- ABDM / HL7 FHIR R4 baseline: spirometry, FeNO, IgE, comorbidities, regimen

### Wearable telemetry

- PPG heart rate, HRV, SpO2, respiratory rate, PEF and smart inhaler events

### Air quality

- Open-Meteo GPS feed: PM2.5, PM10, NO2, SO2, O3, temperature, humidity

### Digital twin engine

- Airway resistance, lung compliance, A-a gradient, autonomic tone
- Risk ensemble with calibrated uncertainty
- 24-hour forward projection of the Exacerbation Risk Index
- What-if sandbox for candidate interventions

### Forecast 8 to 24 h ahead

- Exacerbation and airway-resistance surge before symptoms

### Personalised protocol

- Best of HEPA, proactive inhaler or shifted outdoor time

### Clinician decision support

- Triage, sign-off queue, 30-day GINA summary PDF

**Guardrails by design**

Deterministic safety rails sit between the model and the user: acute-distress bypass to 108 / 112, GINA override, output sanitizer and a human approval queue. The system assists clinicians and patients. It never diagnoses and never changes prescriptions on its own.

Demo persona in the video: a 28-year-old male with moderate persistent asthma. All data is synthetic.

# System architecture: five layers, left to right

## PulmoTwin | System Architecture

Multimodal cardiopulmonary digital twin for proactive adverse-event forecasting and personalised clinical intervention

Digital Twin Challenge 2026 - Happiest Health Summit
Chronic asthma / cardiopulmonary care, India
Data flows left to right

### 1 DATA SOURCES
Multimodal inputs

**ABDM / HL7 FHIR R4 EHR**
*digital_twin/ehr_connector.py*
- Longitudinal static baseline record
- Spirometry (FEV1, personal-best PEF), FeNO, IgE, blood eosinophils
- Diagnosis (ICD-10), comorbidities, regimen, past exacerbations
- Synthetic data modelled on Synthea / MIMIC-IV, zero PII

**Wearable IoT telemetry**
*digital_twin/wearable_sim.py*
- Continuous PPG: heart rate, HRV rMSSD, SpO2, respiratory rate
- Peak flow (PEF) and FEV1 readings
- Smart inhaler actuation over BLE (puff count and timing)
- Simulated sensor stream in the prototype

**Atmospheric telemetry**
*Open-Meteo air quality + weather APIs*
- Live GPS sync with 48-hour hourly forecast
- US AQI, PM2.5, PM10, NO2, SO2, O3, CO2
- Temperature and humidity
- City sensor grid feeds the physician view

**Patient input**
*web_ui/index.html*
- Bilingual English / Hindi intake, voice and text
- Symptom and PEF diary: wheeze, cough, chest tightness, reliever puffs
- 3-step assessment wizard, manual entry fallback
- Free text is safety-scanned before any model sees it

### 2 INGESTION AND BACKEND
Fusion, API, storage

**Connectors and adapters**
*ehr_connector.py, wearable_sim.py*
- FHIR R4 bundle mapped to the internal patient schema
- Wearable and inhaler events normalised into time series
- Open-Meteo data fetched per GPS location
- data_generator.py builds the synthetic cohort

**Preprocessing and fusion**
*preprocess.py, utils.py*
- Cleans and encodes clinical, sensor and air-quality features
- Fuses the three streams into one multimodal feature vector
- Fitted pipeline saved in preprocessing.pkl

**Flask REST API**
*app.py*
- Flask 3.1 served by Gunicorn
- /api/digital-twin/* endpoints, including simulate
- Token authentication and IDOR guards
- Serves the patient portal and physician console

**Persistence**
*SQLite via Flask-SQLAlchemy*
- Profiles, sensor readings, diary entries, agent plans
- Tamper-evident audit trail of every action
- Stored locally, never sold or shared without consent

### 3 DIGITAL TWIN ENGINE
Simulate, predict, test

**Physiological twin**
*digital_twin/twin_engine.py*
- Airway resistance: Raw = 1.8 / PEF_ratio^1.6
- Lung compliance: Cdyn = 70 x PEF_ratio - FeNO/10
- Alveolar-arterial O2 (A-a DO2) gradient
- Autonomic sympathetic tone from HRV rMSSD
- Exacerbation Risk Index (ERI) from 0 to 1

**Multimodal risk ensemble**
*models.py*
- Soft-voting stack: Logistic Regression (w1), Random Forest, 100 trees (w2), Gradient Boosting, 150 stages (w2)
- Accuracy 73.0%, Brier 0.1393 on N=300 held-out
- Shannon-entropy uncertainty rejects low-confidence output

**24-hour forward projection**
*twin_engine.py (ODE kinetics)*
- Differential kinetics of airway inflammation
- Forecasts Raw surges and exacerbations 8 to 24 h before onset
- ERI tiers: green controlled, yellow caution, red critical
- Feeds the 48-h safe-window timeline

**Counterfactual what-if sandbox**
*digital_twin/counterfactual.py*
- Tests interventions in silico before acting
- Indoor HEPA filter: PM2.5 down 75%, risk down 45.2%
- Proactive SMART inhaler: risk down 66.7%, ERI 0.84 to 0.28
- Shifts evening run to the 06:30-08:30 window

### 4 SAFETY RAILS
Deterministic, no black box

**Acute distress interceptor**
*input guard, runs first*
- 61 distress phrases and 32 physical signs
- English, Hindi and Hinglish, negation-aware
- Bypasses the model: instant SOS, dial 108 / 112
- Also fires when PEF drops below 50% of best

**GINA clinical override**
*GINA Step-5 rules*
- Daily symptoms or frequent night waking force High Risk
- High-risk recall rises 72.09% to 82.95% (107 of 129)
- Deliberate bias: a false alarm costs less than a miss

**Output sanitizer**
*post-generation filter*
- Regex check on every agent reply
- Blocks dosage changes and diagnostic claims
- Swaps violations for a clinical fallback disclaimer

**Zero-trust approval queue**
*human in the loop*
- Agent never sends SMS or edits external systems itself
- Every action stays PENDING until a human approves
- Clinician sign-off queue for physician-side plans

### 5 INTERFACES AND OUTPUTS
Patient and clinician

**Patient portal**
*web_ui/index.html*
- Twin cockpit: live telemetry, organ biomechanics, What-If
- Autonomous agent: 48-h air-quality timeline, voice and text assistant, symptom diary
- Patient assessment: risk score with action plan

**Physician console**
*view-admin*
- Cohort triage table with per-model scores
- KPI cards: cohort, sensor nodes, inferences, alerts
- Clinician sign-off queue and Quick Prescribe (GINA)
- Broadcast advisory to the cohort

**Reports and privacy**
*ReportLab 5.0*
- 30-day GINA consultation summary PDF
- JSON export for data portability
- Permanent account and health-record erasure
- Aligned with DPDP Act 2023 and GDPR principles

**Deployment targets**
*Pages, Render, Vercel, Docker*
- GitHub Pages: in-browser physiology engine
- Render, Vercel, Docker and Procfile for the Flask app
- pytest suite covers guardrails, agent and API

**TECHNOLOGY STACK** Icons are simplified vector marks, not official brand logos

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Python 3.11 core language | Flask 3.1 REST API, Gunicorn | SQLite SQLAlchemy ORM | NumPy, SciPy ODE twin engine | scikit-learn stacking ensemble | ReportLab PDF reports | HL7 FHIR R4 ABDM schemas | Open-Meteo air quality, weather | Docker container, Render | HTML5, ES6 JS web cockpit |

PulmoTwin is an investigational decision-support research prototype, not a certified medical device. In an emergency dial 112 / 108.
Source: github.com/Kabirroy12345/Digital_Twin | MIT license | Metrics from README held-out evaluation (N=300)

### Sources
ABDM EHR, wearables, Open-Meteo and patient input

### Backend
Flask API fuses streams, stores in SQLite

### Twin engine
Physiology, ensemble, projection, what-if

### Safety rails
Four deterministic checks before any output

### Interfaces
Patient portal, physician console, reports

# Two data streams fused with live air quality

### Static: ABDM / FHIR R4 EHR

- Spirometry FEV1, personal-best PEF, FeNO, IgE, eosinophils
- Diagnosis, comorbidities, regimen, past exacerbations
- Module: digital_twin/ehr_connector.py

### Dynamic: wearable stream

- PPG HR, HRV rMSSD, SpO2, respiratory rate, PEF, FEV1
- BLE smart inhaler actuation
- Module: digital_twin/wearable_sim.py

### Atmospheric: Open-Meteo

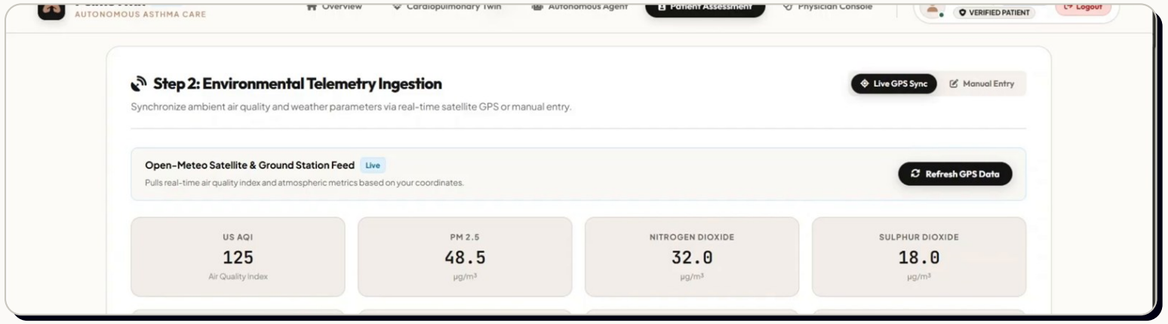
- Live GPS sync, 48-hour hourly forecast
- US AQI, PM2.5, PM10, NO2, SO2, O3, CO2, temperature, humidity
- Bengaluru sensor grid for the physician view

### Patient: bilingual intake

- English and Hindi voice and text
- Symptom and PEF diary, 3-step assessment wizard
- Free text is safety-scanned first

### Privacy-safe by construction

- 100% synthetic data following Synthea and MIMIC-IV clinical distributions, zero PII
- Preprocessing pipeline saved in preprocessing.pkl; dataset described in DATASET_OVERVIEW.md




*Patient assessment, step 2: live GPS air-quality ingestion (demo video)*

### What the video shows

- One click syncs satellite and ground-station data for the patient's coordinates
- Readings shown: US AQI 125, PM2.5 48.5, NO2 32.0, SO2 18.0, humidity 62, CO2 415
- Manual entry fallback when GPS or network is unavailable
- Output feeds the 3-step assessment: guideline screening, telemetry, risk and action plan

# The physiological twin: lungs and autonomic tone in numbers

**Airway resistance**
`Raw = 1.8 / PEF_ratio^1.6`
**2.34 cmH2O/(L/s) in demo**

**Dynamic lung compliance**
`Cdyn = 70 x PEF_ratio - FeNO/10`
**54.3 mL/cmH2O**

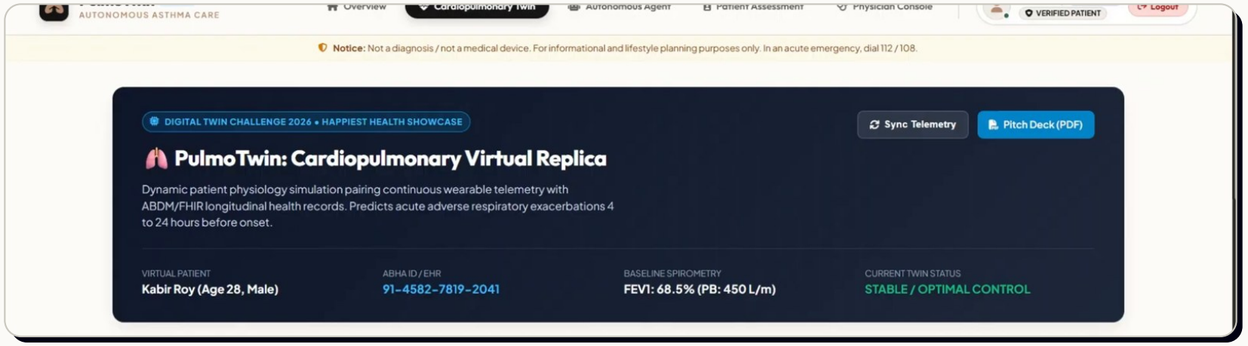
**A-a DO2 gradient**
`Alveolar-arterial oxygen gap`
**13.8 mmHg**

**Autonomic sympathetic tone**
`Derived from HRV rMSSD`
**0.27 (relaxed)**

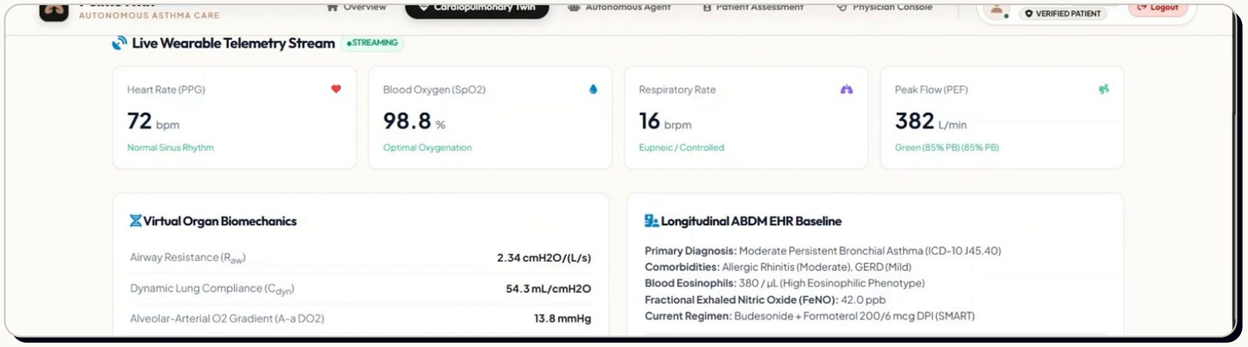
**Exacerbation Risk Index**
`Composite 0 to 1 tier score`
**0.239 stable / optimal**

### 24-hour forward projection

- ODE differential kinetics of airway inflammation drive the forecast
- Surges in Raw are flagged 8 to 24 hours before onset
- ERI tiers: green controlled, yellow caution, red critical




*Cardiopulmonary Twin page: virtual replica header (demo video)*




*Live wearable telemetry, virtual organ biomechanics, ABDM EHR baseline*

### Demo patient values shown in the video

- Vitals: HR 72 bpm, SpO2 98.8%, resp rate 16, PEF 382 L/min; baseline FEV1 68.5%, best PEF 450
- Moderate persistent asthma (ICD-10 J45.40), allergic rhinitis, GERD; FeNO 42 ppb, eosinophils 380/uL

# Risk ensemble: stacked models plus a clinical safety override

| Model | Role | Accuracy | Weighted F1 | Sensitivity |
|---|---|---|---|---|
| **Logistic Regression** | Linear baseline | 71.33% | 0.7117 | 73.64% |
| **Random Forest** | Non-linear trees | 72.33% | 0.7058 | 69.77% |
| **Gradient Boosting** | Sequential boosting | 69.67% | 0.6942 | 71.32% |
| **Stacking ensemble (pure ML)** | Learned collaborative | **73.00%** | **0.7234** | 72.09% |
| **Hybrid system (+ GINA override)** | Clinical safety guardrails | 63.67% | 0.6243 | **82.95%** |

Held-out test set, N = 300 (data/test.csv). Brier score for high-risk probability: 0.1393.

**Logistic Regression**
L2 penalty, weight 1

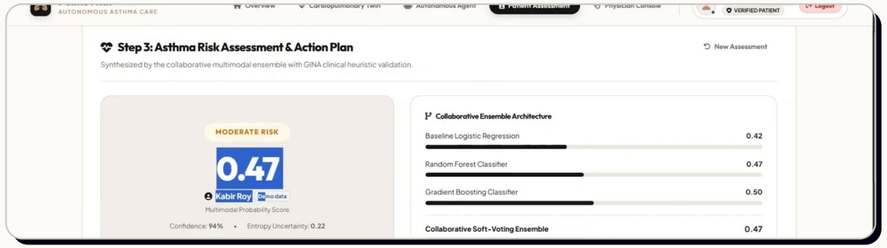
**Random Forest**
100 trees, depth 8, weight 2

**Gradient Boosting**
150 stages, rate 0.05, weight 2

## Why accuracy drops but sensitivity rises

In triage a false alarm costs a rescheduled visit, while a false negative risks a preventable hospitalisation. The GINA Step-5 override intentionally escalates patients with daily symptoms or frequent night waking to High Risk. Sensitivity moves from 72.09% to 82.95% (107 of 129 acute cases detected) at the price of 9.3 points of accuracy.

Uncertainty: Shannon entropy H = -sum(p log2 p) rejects low-confidence predictions.




*Patient assessment, step 3: risk score and ensemble view*

## One patient, end to end (demo)

- Baseline LR 0.42, Random Forest 0.47, Gradient Boosting 0.50
- Collaborative soft-voting ensemble: 0.47, moderate risk
- Confidence 94%, entropy uncertainty 0.22
- Output: clinical care directive, activity plan, respiratory barrier protection, SOS broadcast

# Counterfactual what-if sandbox: test the plan before living it

Clinicians and patients can switch interventions on and off and see the 24-hour exacerbation risk change in the twin, before anything is tried on the patient.

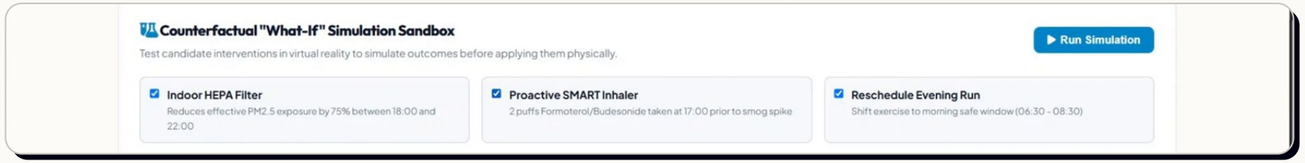
**Patient baseline**
Projected peak ERI 0.84, critical decompensation
0.84 RED

**A: Indoor HEPA filter**
PM2.5 exposure cut 75% from 18:00 to 22:00. Risk reduction 45.2%
0.46 YELLOW

**B: Proactive SMART inhaler**
2 puffs formoterol/budesonide at 17:00. Risk reduction 66.7%
0.28 GREEN

**Optimal protocol selected: HEPA + proactive inhaler + shifted run**

Video result: 66.7% reduction in peak exacerbation risk, ERI falls from 0.84 red to 0.28 green, airway resistance stays below 3.0.




*Cardiopulmonary Twin page: three selectable interventions (demo video)*

### Indoor HEPA filter

- Cuts effective PM2.5 exposure for the evening peak

### Proactive SMART inhaler

- Pre-emptive controller puffs before the smog spike

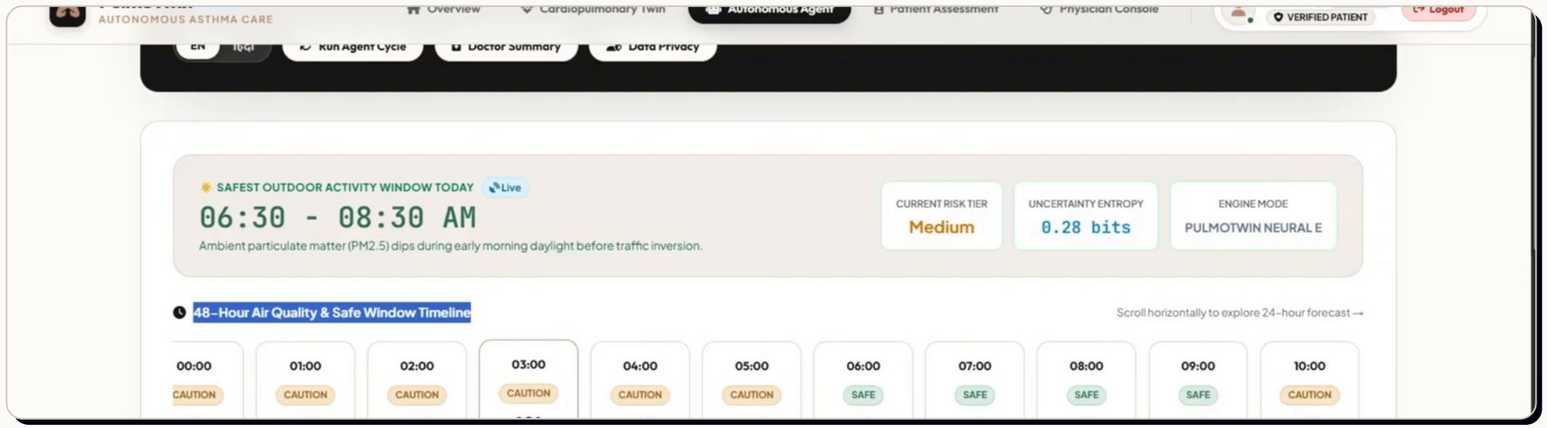
### Reschedule evening run

- Move exercise to the 06:30 to 08:30 safe window

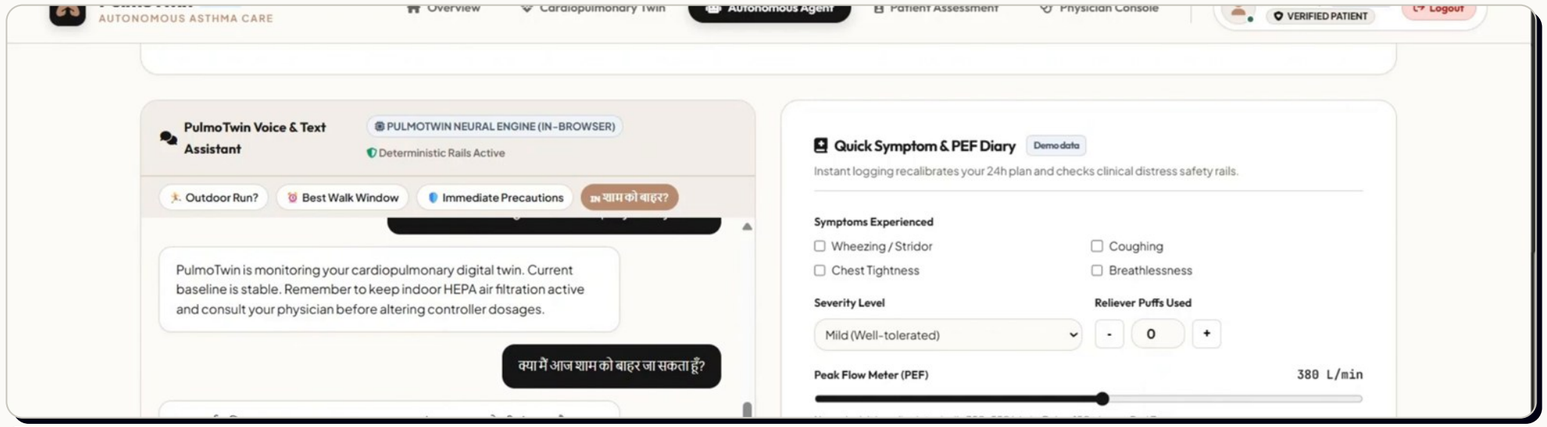
### Why it matters

- Moves from generic advice to a ranked, personalised protocol
- Safe to explore: simulation only, nothing is dispatched
- Module: digital_twin/counterfactual.py

# Autonomous patient agent: bilingual, proactive, always asks first




*Autonomous Agent: safest outdoor window and 48-hour air-quality timeline*




Portal pages: Overview | Cardiopulmonary Twin | Autonomous Agent | Patient Assessment | Physician Console

### Plans the day around air

Safest outdoor window (06:30 to 08:30 in the demo), hourly 48-hour AQI timeline and a preventive checklist

### Speaks the patient's language

One-click English / Hindi UI, voice and text assistant on a deterministic neural engine

### Logs and learns

Quick symptom and PEF diary recalibrates the plan; risk attribution and decision log explain each action

### Asks before acting

Proposed actions appear under Actions Requiring Your Approval

# Safety rails and privacy: deterministic, auditable, human in the loop

### Acute distress interceptor

61 phrases and 32 physical signs in English, Hindi and Hinglish, negation-aware. Bypasses the model and calls 108 / 112.

### GINA clinical override

Step-5 rules escalate daily symptoms or frequent night waking to High Risk.

### Output sanitizer

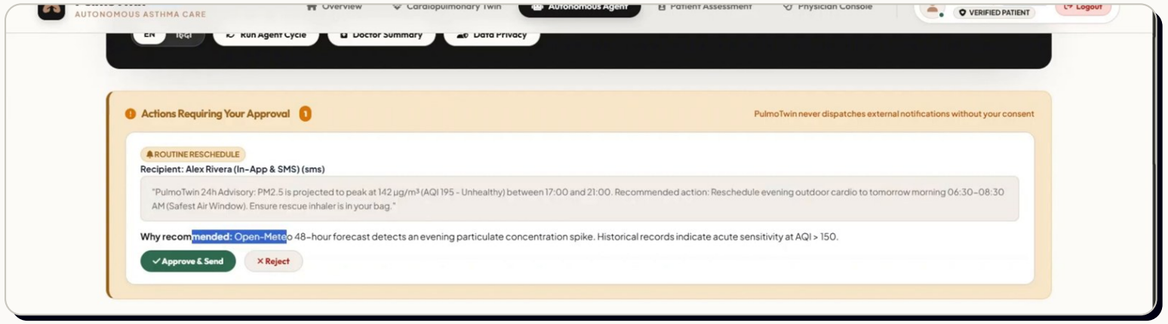
Regex filter blocks dosage changes such as extra puffs and diagnostic claims, then substitutes a safe disclaimer.

### Zero-trust approval queue

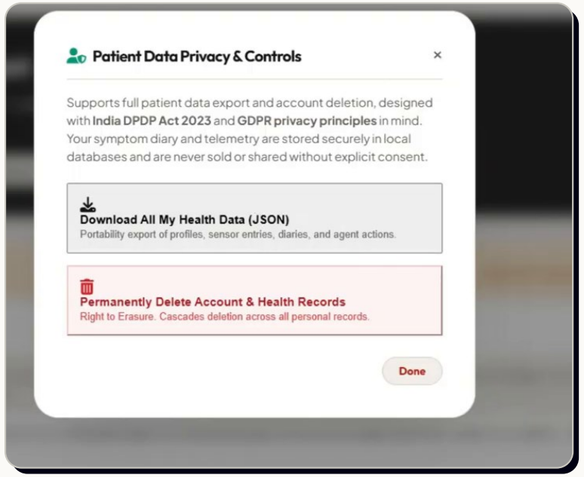
Nothing is sent automatically. Actions stay PENDING until a human approves.

## Data protection

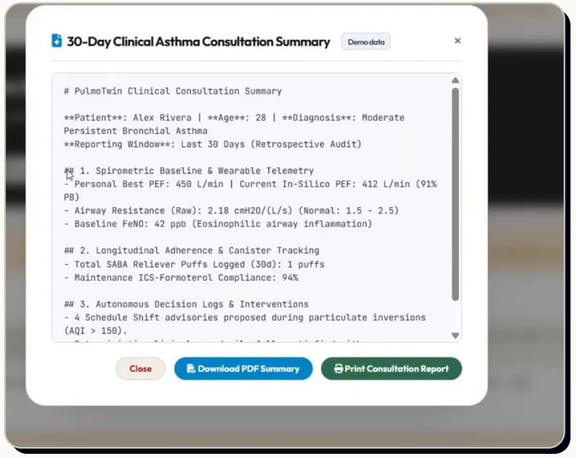
- Designed with India's DPDP Act 2023 and GDPR principles in mind
- One-click JSON export of profile, sensor series, diary and agent actions
- Permanent delete of account and health records with cascading erasure
- Tamper-evident audit trail in SQLite; token auth and IDOR guards on every endpoint
- Unlike a black-box LLM, unsafe advice is blocked in code, not by prompt




*Agent: Actions Requiring Your Approval with an approve / reject choice*



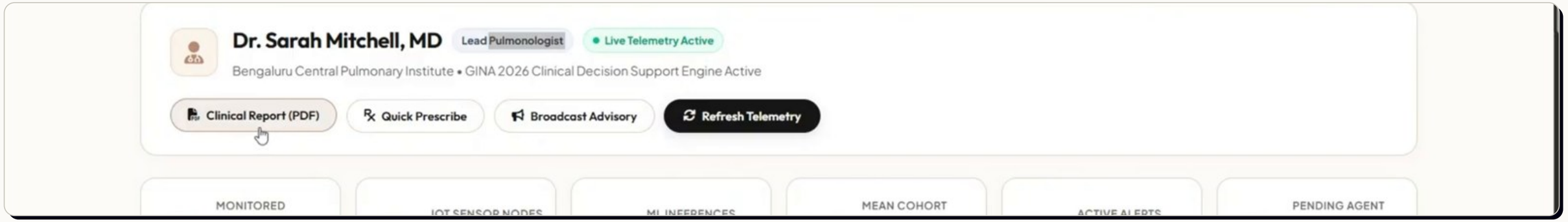

*Patient Data Privacy & Controls*

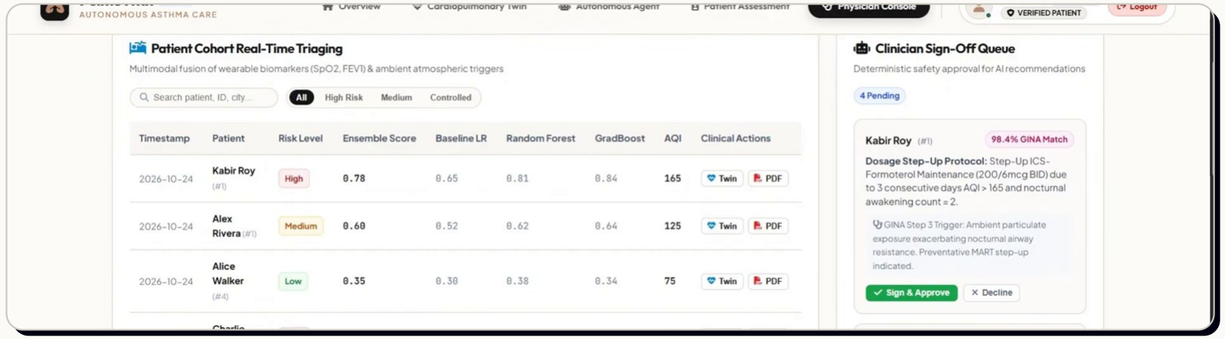



*30-Day Clinical Consultation Summary*

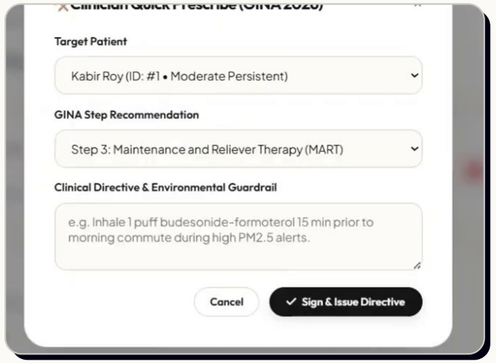
Result: 82.95% high-risk sensitivity and zero automatic dispatches, with every action traceable.

# Physician console: triage a cohort in one screen



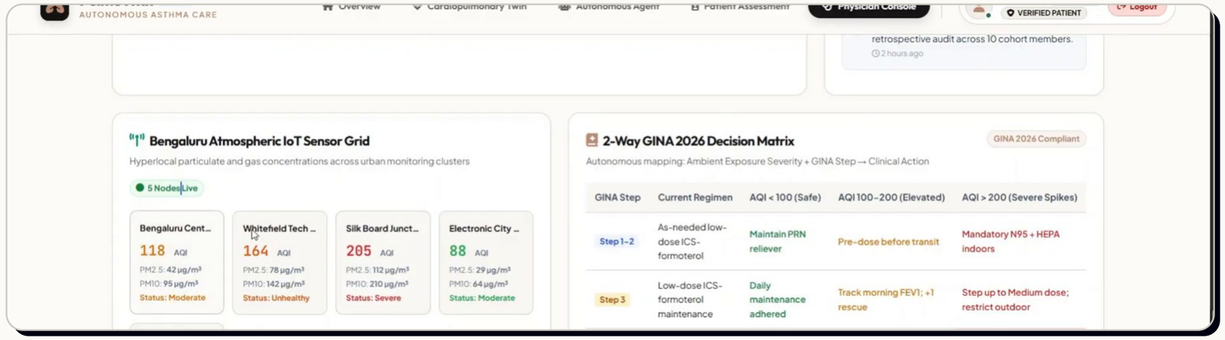

*Cohort triage table with per-model scores, plus the clinician sign-off queue*




*Quick Prescribe (GINA 2026)*

## Console at a glance

- KPIs: cohort 10, sensor nodes 26, ML inferences 28, mean risk 0.42, alerts, pending plans
- Priority list by risk level with Twin and PDF buttons
- Broadcast advisory to the cohort




*Bengaluru atmospheric IoT sensor grid and 2-way GINA 2026 decision matrix*

**Triage**
High, medium, controlled; ensemble, LR, RF and GB scores plus AQI per patient

**Sign off**
Approve or decline each AI plan with a one-line rationale

**Prescribe**
Select the GINA step and issue a signed directive

ENGINEERING

# Technology stack, quality and deployment

| Layer | Technologies |
|---|---|
| **Twin engine** | Python 3.11, NumPy, SciPy, ODE kinetics |
| **Machine learning** | scikit-learn: LR + RF + GB stacking, entropy, Brier |
| **Backend** | Flask 3.1, Flask-SQLAlchemy, SQLite, Gunicorn |
| **Standards** | HL7 FHIR R4, Ayushman Bharat Digital Mission |
| **Safety** | Deterministic regex rails, GINA Step-5 protocols |
| **Reporting** | ReportLab 5.0 PDF, JSON data portability |
| **Frontend** | HTML5, vanilla ES6 JS, CSS3, Warm Ivory design system |
| **Atmospheric data** | Open-Meteo air quality and forecast APIs |

Metrics and stack as stated in the project README.

### Quality

- Automated pytest suite for guardrails, agent, twin engine, API and offline frontend; QA_REPORT.md and AUDIT.md in the repo

### Deployment

- GitHub Pages live cockpit with in-browser physiology engine; Render, Vercel, Docker and Procfile for the Flask app

### Repository map

- agent, digital_twin, web_ui, n8n workflows, research, results, scripts, tests, docs, slideshow, figures

### Reproduce it in 3 commands

```
git clone github.com/Kabirroy12345/Digital_Twin
pip install -r requirements.txt
python app.py   # http://localhost:7860
```

License: MIT. Docs also include LIMITATIONS.md and RESEARCH_RESULTS.md.

WRAP-UP

# Summit alignment, impact and what comes next

| Summit requirement | PulmoTwin implementation |
|---|---|
| **Dynamic virtual replica** | Raw, Cdyn, autonomic tone, 24 h projection |
| **Wearables integration** | PPG telemetry stream, BLE smart inhaler |
| **Longitudinal EHR pairing** | ABDM / HL7 FHIR R4 baseline |
| **Predict adverse events** | Forecast 8 to 24 h before onset |
| **Personalise treatment** | What-if sandbox, ranked protocols |
| **Indian realities** | English / Hindi UI, 108 / 112 SOS, urban PM2.5 |

## Proposed next steps

- Validate on real, consented ABDM-linked cohorts with clinician review
- Replace the simulated wearable stream with live BLE and PPG devices
- Pursue the CDSCO regulatory pathway before any clinical use
- Extend the same twin pattern to COPD and other cardiopulmonary conditions

## Impact we are aiming for

- Fewer emergency nebulisations and ICU admissions through earlier action
- Patients who plan around air quality instead of reacting to it
- Physicians who walk in with 30 days of objective data

## Team PulmoTwin

Team lead: Kabir Roy

**LIVE COCKPIT**
kabirroy12345.github.io/Digital_Twin

**SOURCE CODE (MIT)**
github.com/Kabirroy12345/Digital_Twin

**FULL DEMO VIDEO (23 MIN)**
youtube.com/watch?v=i7NJstq8_ZE

## Thank you

*Questions and a live walkthrough are welcome.*

Research prototype, not a medical device. Emergency: 112 / 108.